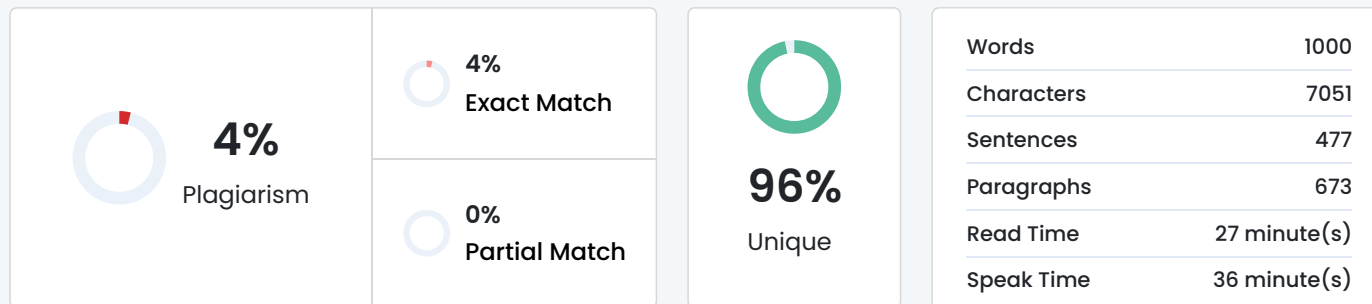


Plagiarism Scan Report



Content Checked For Plagiarism

SIMATS ENGINEERING

Saveetha Institute of Medical and Technical Sciences

Chennai – 602105

A COURSE ASSIGNMENT REPORT

ON

Design and Develop an Intelligent Flood Disaster Detection
and Emergency Alert Prediction System Using Deep Learning

Submitted in partial fulfilment for the Course of

Course Code – CSA0403 / MLA04 – Deep Learning

to the award of the degree of

BACHELOR OF ENGINEERING

IN

COMPUTER SCIENCE AND ENGINEERING

Submitted by

Shaik Rafiqhuddin (192525129)

Kothakota Rakesh (192525075)

A. Santhosh Kumar Reddy (192525269)

Under the Supervision of

Dr. A. M. ARUL RAJ

Professor, Department of Computer Science and Engineering

Date of Submission – September 2026

SDG Mapping: SDG 11 – Sustainable Cities | SDG 13 – Climate Action | SDG 9 – Innovation | SDG 3 – Health

Bloom's Taxonomy Level: L6 – Create | Course Outcomes: CO2, CO3

CSA0403 – Intelligent Flood Detection System | Deep Learning

Page 2 of 23 | SIMATS Engineering | Sep 2026

Table of Contents

Chapter No. Title Page

1 Problem Statement 4

2 Aim and Objectives 6

3 Requirements / Environment and Tools Used 7

4 Design of the Module 9

5 Proposed Solution 11

6 Algorithm 13

7 Pseudocode 14

8 Flowchart 15

9	Implementation	16
10	Source Code	18
11	Test Cases	22
12	Expected / Actual Results	23
13	Execution Screenshots / Output	24
14	Analysis Result Charts	25
15	Discussion and Limitations	27
16	Conclusion	28
17	Future Enhancement	29
18	Individual Contribution of Group Members	30
19	References	31
List of Tables		
S.No	Table Name	Table No. Page
1	Input Parameters and Data Sources	Table 3.1 8
2	Dataset Description and Class Distribution	Table 3.2 8
3	Model Hyperparameters	Table 5.1 12
4	Performance Metrics Comparison	Table 12.1 23
5	Test Case Summary	Table 11.1 22
6	Architecture Comparison (AlexNet vs ResNet)	Table 14.1 26
List of Figures		
S.No	Figure Name	Fig No. Page
1	System Architecture Diagram	Fig 4.1 9
2	CNN Architecture for Flood Images	Fig 4.2 10
3	System Flowchart	Fig 8.1 15
4	Training & Validation Accuracy/Loss Curves	Fig 14.1 25
5	Confusion Matrix	Fig 14.2 25
CSA0403 – Intelligent Flood Detection System Deep Learning		
Page 3 of 23 SIMATS Engineering Sep 2026		
S.No	Figure Name	Fig No. Page
6	Performance Metrics Bar Chart	Fig 14.3 26
7	Risk Distribution Pie Chart	Fig 14.4 26
8	Architecture Comparison Chart	Fig 14.5 26
1	Problem Statement	Chapter

1.1 Introduction

Floods are among the most frequent and destructive natural disasters worldwide, causing significant loss of life, displacement of communities, damage to infrastructure, and economic disruption. In India and many developing regions, flood monitoring still relies heavily on manual observation, traditional meteorological analysis, and delayed ground reporting. These conventional approaches suffer from late detection of rising flood risk, difficulty in processing large volumes of multi-source environmental data, delayed emergency response, and limited ability to recognise complex spatial-temporal patterns that precede flood events.

A disaster management authority therefore requires an Intelligent Flood Disaster Detection and Emergency Alert Prediction System that can continuously analyse heterogeneous data streams—rainfall intensity, river/water level, temperature, humidity, geographical attributes, historical flood records, and satellite or surveillance imagery—and produce timely, graded risk alerts (Normal, Low Risk, Moderate Risk, High Risk, Severe Flood Risk). Deep Learning offers powerful tools for this task through representation learning, hierarchical feature extraction, and end-to-end classification.

1.2 Limitations of Existing Systems

- Heavy dependence on manual observation and delayed human reporting.
- Inability to process high-volume, multi-modal environmental and image data in near real time.
- Traditional statistical weather models fail to capture non-linear interactions among rainfall, soil saturation, topography and drainage.
- Absence of automated, graded emergency-alert generation linked to predicted risk severity.

- Limited generalisation across different geographic basins and climate regimes.
- Lack of integration between structured sensor data and unstructured satellite/camera imagery.

1.3 Problem Definition

Design, develop and evaluate a Deep Learning-based system that (a) ingests structured environmental parameters and/or flood-related images, (b) learns discriminative representations of normal versus flood-prone conditions, (c) classifies the instantaneous flood-risk level into discrete categories, and (d) triggers understandable emergency alerts for disaster-response authorities and at-risk communities. The system

CSA0403 – Intelligent Flood Detection System | Deep Learning

Page 4 of 23 | SIMATS Engineering | Sep 2026

must demonstrate the application of representation learning, appropriate neural-network width/depth, activation functions (ReLU, Leaky ReLU, Softmax, etc.), autoencoders where beneficial, CNN components (convolution, pooling, parameter sharing, regularisation), and a justified comparison of architectures such as AlexNet and ResNet.

1.4 Stakeholders and Scope

- Primary users: Disaster Management Authorities, State Emergency Operation Centres, National Disaster Response Force units.
- Secondary users: Local government bodies, meteorological departments, community early-warning networks.
- Scope: Offline and near-real-time batch prediction from historical/sensor/image data; graded risk classification; alert generation. Real-time continuous IoT streaming and mobile push notifications are identified as future enhancements.
- Out of scope: Hydraulic modelling of river discharge, detailed inundation mapping, or autonomous control of flood gates.

🔑 Key Insight: Early, accurate, and graded flood-risk prediction can substantially reduce loss of life and property

by giving authorities and communities actionable lead time—directly supporting SDG 11 (Sustainable Cities) and

SDG 13 (Climate Action).

2 Aim and Objectives Chapter

2.1 Aim

To design, implement and evaluate an Intelligent Flood Disaster Detection and Emergency Alert Prediction System using Deep Learning techniques that can identify patterns associated with flood conditions from multi-source environmental data and images, classify flood-risk severity, and support timely emergency alerting.

2.2 Objectives

1. Analyse the limitations of traditional flood monitoring and clearly define functional and non-functional requirements of the proposed system.
2. Identify suitable multi-modal datasets (structured environmental parameters and/or flood imagery) and perform rigorous data cleaning, normalisation, encoding and augmentation.
3. Apply representation learning and feature-extraction techniques appropriate to the data modality (tabular features or CNN-based hierarchical features).
4. Design a Deep Learning architecture (CNN / hybrid CNN-MLP / ResNet-based transfer learning) with justified choices of depth, width, activation functions, pooling, and regularisation.

CSA0403 – Intelligent Flood Detection System | Deep Learning

Page 5 of

Matched Source

Similarity 1%

Title: Saveetha Institute of Medical and Technical Sciences ...

Saveetha Institute of Medical And Technical Sciences (SIMATS) , Tamil Nadu,India is one of the top Private not for Profit universities in Tiruvallur, India. It ...Read more

[Check](#)

Similarity 1%

Title:[The Award of the Degree of PhD on the Basis of Published ...](#)

This report relates to the award of the degree of PhD to a candidate whose thesis consists entirely or predominantly of refereed and published articles in ...Read more

[Check](#)

Similarity 1%

Title:[Welcome to www.4training.net - 4training](#)

This website wants to serve you and make your job easier by equipping you with simple training worksheets in many languages.

<https://google.com/goto?url=CAESagHrOzAVb-USxxGQGYTybfz0R2ZDzmTH8KAL7RVumFlfN8sGflaQBpGo7lzO4mecdm0ET6bLQkh0GZjnXtAgg5MYv2C2JSqs4JyrNfP3oesmil3v6G8yTdBUV0unFon3knupty74eKvyDa0>
